SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Problem-Solution Fit

Date	03 july 2026
Team ID	xxxxxx
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS]

- Farmers
- Agricultural Officers
- Agricultural Students
- Agronomists
- Smart Farming Organizations

6. CUSTOMER LIMITATIONS

- Limited technical knowledge
- Poor internet connectivity
- Budget constraints
- Limited access to experts
- Low awareness of AI tools

5. AVAILABLE SOLUTIONS

- Traditional farming knowledge
- Agricultural experts
- Government advisory centers
- Internet articles
- Weather reports

2. PROBLEMS / PAINS

- Difficulty selecting the best crop.
- Incorrect crop selection causes financial loss.
- Farmers depend on traditional experience.
- Soil nutrients are not properly analyzed.
- Low agricultural productivity.

9. PROBLEM ROOT

- Lack of scientific crop selection.
- No proper soil analysis.
- Climate uncertainty.
- Limited access to agricultural experts.
- Traditional decision-making methods.

7. BEHAVIOR

- Consult local farmers.
- Search information online.
- Visit agriculture offices.
- Use previous farming experience.

3. TRIGGERS TO ACT

- Start of farming season.
- Soil testing completed.
- Need to improve crop yield.
- Reduce farming risks.
- Increase profit using AI recommendations.

4. EMOTIONS

BEFORE:
- Confused
- Worried
- Uncertain
- Fear of crop failure

AFTER:
- Confident
- Informed
- Satisfied
- Better decision making

10. YOUR SOLUTION

OptiCrop is an AI-powered crop recommendation system that analyzes soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall using a trained Random Forest Machine Learning model to recommend the most suitable crop with high accuracy. Future enhancements include SHAP Explainability, Weather API integration, Confidence Score, What-if Simulator, and Fertilizer Recommendation.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE
- Website
- Mobile Applications
- Government Agriculture Portals
- YouTube

OFFLINE
- Agriculture Officers
- Local Farmers
- Soil Testing Centers
- Farming Workshops